

Md Rahman

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EDUCATION

City College of New York, Grove School of Engineering

New York, NY

B.E. in Mechanical Engineering

Expected May 2029

- GPA: 4.00/4.00 — Dean's List — S. Jay Levy Fellowship (2026–2027) — Celia Portnoy Memorial Scholarship
- Coursework: Engineering Mechanics I/Statics (A+), Calculus III (A), Computer-Aided Drafting (A+), University Physics I & II (A+)

Stuyvesant High School — New York, NY — High School Diploma, 2025

TECHNICAL SKILLS

CAD & Fabrication: SolidWorks, Onshape, FDM 3D Printing (Bambu Lab P1P), Laser Cutting, Machining

Analysis & Programming: MATLAB, Python (NumPy, Matplotlib), Arduino (Embedded C), PID Control Tuning

Testing & Hardware: Test Fixture Design, Bench Load Testing, Sensor Integration, BOM Development

ENGINEERING EXPERIENCE

CarbonCLAIR — Environmental Technology Startup

New York, NY

Engineering Intern

Summer 2026 – Present

- Designed a vacuum chamber for a 13.56 MHz capacitively coupled RF plasma reactor running argon at 30–150 mTorr, sizing bore and electrode gap from Debye length and mean-free-path calculations and verifying 8–25x buckling safety factors under 1 atm external differential; documented in a 31-page design report and 23-figure drawing set.
- Sourced a procurement BOM of verified part numbers across 10+ vacuum suppliers, catching two catalog RF feedthroughs sold under a vendor's RF product line but rated at 450 kHz rather than the required 13.56 MHz; cost-modeled a \$19K staged starter build against a \$75K+ research-grade configuration.
- Built the droplet-detection subsystem for an electrohydrodynamic droplet test rig: modeled and 3D-printed a fixture carrying an Arduino Uno and laser-photodiode beam-break sensor, wrote the detection firmware, and validated real-time frequency logging against Photron high-speed imaging under kV-scale actuation.

BioWearLab — Grove School of Engineering

New York, NY

Undergraduate Research Assistant, Mechanical Engineering

2026 – Present

- Engineered BCC lattice metamaterials in SolidWorks for additively manufactured TPU-graphene pressure sensors, optimizing unit-cell geometry across 8+ iterations to hit 0–50 N and 0–100 N gait-force ranges within 5% of target stiffness.
- Designed and manufactured 5+ test fixtures, compression mounts, and loading rigs on a Bambu Lab P1P (+/-0.2 mm verified via digital calipers), enabling repeatable sensor characterization under compression, shear, and cyclic loading.
- Built automated Python and MATLAB pipelines analyzing 500+ data points per test cycle, generating force-displacement plots and summary tables that cut weekly reporting time roughly 40%.

FRC Team 694 (StuyPulse)

New York, NY

Mechanical Lead

2024 – 2025

- Designed a custom drivetrain and two-stage gearbox in SolidWorks/Onshape for a 120 lb competition robot, cutting mass 12% (4.8 lbs) via plate pocketing and ribbed geometry; verified stiffness with hand calculations and bench load testing.
- Integrated FDM-printed intake, manipulator, and bracket assemblies with PID control loops on encoder and IMU feedback, tuning gains over 15+ iterations to reach +/-3 cm autonomous accuracy across a 5 m traverse.

LEADERSHIP & OPERATIONS

Khan's Tutorial | Branch Manager | Queens, NY

2024 – Present

- Oversee operations, enrollment, and instruction across 6 branches serving 180+ students and 8 instructors; closed 30+ enrollments totaling \$30,000+ in revenue and cut missed sessions 20% via attendance tracking.

MIST New York | Operations Lead | New York, NY

2024 – Present

- Directed day-of logistics for a 1,150-attendee tournament at Adelphi University, coordinating 80+ volunteers across 30+ competition events delivered 8% under budget.